

VADALA PAVAN KUMAR

DevOps | Site Reliability | DevSecOps

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PROFESSIONAL SUMMARY

DevSecOps / Site Reliability Engineer with around 5.7+ years of experience designing CI/CD pipelines, automating hybrid infrastructure (on-prem + AWS), and supporting production systems across DEV, QA, UAT, and PROD. Skilled in Bash, Python, and Ansible automation that reduced manual deployment effort and improved release frequency. Experienced in incident response, root cause analysis (RCA), and zero-downtime rollback strategies. Hands-on with AWS (EC2, VPC, IAM, S3, RDS, ALB, Route 53, CloudWatch) and working knowledge of Azure. Proven ability to bridge development and operations by resolving deployment-level backend issues and optimizing database performance (PostgreSQL, Pg Bouncer).

TECHNICAL SKILLS

- **Cloud & Infrastructure:** AWS (EC2, VPC, IAM, S3, RDS, ALB, ECR, Route 53, CloudWatch), Azure (basic production experience), Hybrid Infrastructure (On-Prem + Cloud)
- **CI/CD & Build Tools:** Jenkins, GitHub Actions, GitLab CI/CD, Bitbucket Pipelines, Release Management, Branch Protection & Governance
- **Containers & Orchestration:** Docker, Kubernetes (hands-on practice)
- **Automation & Scripting:** Bash, Python, Ansible, Cron-based Automation
- **Monitoring & Observability:** Prometheus, Grafana, CloudWatch, Alerting & Notification Systems
- **Reliability & Operations:** Incident Management, Root Cause Analysis (RCA), Rollback Strategies, High Availability, Zero-Downtime Deployments
- **Databases & Backend:** MySQL, PostgreSQL, MongoDB, Pg Bouncer (Connection Pooling & Optimization), Node.js, Python
- **Servers & OS:** Linux (Ubuntu, RHEL, CentOS), Nginx, Apache, Tomcat
- **Security & Access Control:** IAM Policies, RBAC, Secure Access Control for Production Systems
- **Version Control & Code Quality:** Git, GitHub, GitLab, SonarQube, Nexus
- **Other Tools:** Jira, Postman

PROFESSIONAL EXPERIENCE

DevOps Engineer / Site Reliability Engineer — Varun Group

April 2021 – Present

Projects: Varun Management System (VMS), V-Promise, HRMS, EAM, VMS-Audit, MMS, Cosec-Backend (3-tier applications)

- Designed and implemented CI/CD pipelines using Jenkins, GitHub Actions, GitLab CI/CD, and Bitbucket Pipelines across DEV, QA, UAT, and PROD environments, standardizing releases across [Number] applications.
- Automated deployments using Bash and Ansible, enabling branch-based deployment workflows and reducing manual release effort by [60%].
- Built and managed hybrid infrastructure (On-Prem + AWS), including migration support and environment setup for multiple production applications.
- Implemented release governance with branch protection, approval workflows, tagging strategies, and rollback mechanisms, improving release consistency and reducing failed deployments.
- Developed cron-based automation for health checks, database backups, cleanup tasks, and alert notifications, improving system uptime and reducing manual monitoring effort.
- Led production incident response, performing root cause analysis (RCA) and executing Git-based rollback strategies to restore service with minimal downtime.
- Deployed and supported multi-language applications built with Node.js, Python, PHP, Java, React, and Flutter across DEV through PROD environments.
- Integrated SonarQube and Nexus into the CI/CD pipeline for automated code quality checks and artifact management.
- Implemented RBAC and secure access controls for production systems, strengthening the security posture across environments.

- **Key Achievements:** Reduced deployment time by transitioning from manual releases to automated CI/CD pipelines; increased deployment frequency in DEV environments to accelerate testing cycles; achieved zero-downtime deployments through rollback strategies; improved system reliability via proactive monitoring and alerting.

Projects: JPF, VR-Garage, Clokam, HRMS, EAM

- Managed AWS infrastructure including EC2, ALB, Auto Scaling, S3, IAM, RDS, and Route 53 to support production workloads.
- Configured secure IAM roles and access policies for developers and services, enforcing least-privilege access.
- Built and maintained CI/CD pipelines using Jenkins and GitHub Actions to automate build, test, and deployment stages.
- Supported Docker-based application deployments across development and production environments.
- Configured monitoring using CloudWatch, Prometheus, and Grafana dashboards to track system health and performance.
- Set up email-based alerting systems for failures and system anomalies, enabling faster incident response.
- Handled production incidents end-to-end, ensuring high availability of critical applications.
- **Key Contributions:** Designed monitoring dashboards and alerts for proactive failure detection; supported infrastructure scaling and networking setup, including NAT gateways and public/private subnet architecture.

DevOps Intern — Varun Motors Pvt. Ltd., Visakhapatnam

Jan 2021 – March 2021

- Worked hands-on with AWS services including EC2, VPC, IAM, and Linux servers in a production-support capacity.
- Assisted in CI/CD pipeline setup using Jenkins to support application build and deployment workflows.
- Supported application deployments and environment configurations across development environments.
- Built foundational DevOps skills through hands-on implementation and self-directed learning.

PROJECTS

- **Zero-Downtime Deployment System** — Designed deployment strategies with rollback support, ensuring continuous service availability during releases.
- **Database Automation & Migration Scripts** — Automated DEV → QA → UAT → PROD migrations; built backup and restore systems and implemented full database recovery workflows.
- **Safety Application (In Progress)** — Developing an AI-assisted safety system with backend integrations and automation.
- **AI/LLM-Integrated Report Generator (Personal/Hobby Project)** — Built a small application integrating LLMs to automate report generation, exploring prompt design and backend integration for AI-driven output.
- **Personal Portfolio** — Hosted a static portfolio website using GitHub Pages.

KEY ACHIEVEMENTS

- Restored critical production databases after accidental deletion using backup and recovery strategies.
- Implemented zero-downtime deployment pipelines with automated rollback mechanisms.
- Automated deployment and database workflows, significantly reducing manual operational effort.
- Optimized PostgreSQL performance using Pg Bouncer connection pooling and configuration tuning.

CERTIFICATIONS & EDUCATION

- AWS Certified Developer – Associate
- DevOps Hands-On Training
- MCA — Vignan's University, Guntur

ADDITIONAL SKILLS

- AI-Assisted Development: Use ChatGPT, GitHub Copilot, Claude, and Gemini for debugging, automation, log analysis, and CI/CD improvements without exposing sensitive data.
- AI/LLM Integration (Hobby): Personal interest in integrating LLMs into small applications, including a report-generation tool, for hands-on exploration of prompt design and AI-driven automation.